TOPIC 03 · INTERACTIVE DASHBOARD
LAB 12 THEME · OPTIMISATION, REPORTING & SAFE SHARING

Lab 12 — Optimise the Hiring Decision and Share Results Safely

Maps to	LO3: use Gemini's optimisation problem solver to choose hires under constraints, verify the solution with formulas, export a recommendation to Google Docs, use Drive files as context and share HR results responsibly
Duration	30 minutes
Tools	Google Sheets · Gemini optimisation problem solver · Export to Docs · Drive files as context · sharing · OrchidBay_Hiring_Decision.xlsx
Mock data	data/OrchidBay_Hiring_Decision.xlsx
Course	Gemini for Google Sheet (C817) · v2.0
Prompts	https://tertiarycourses.github.io/C817-Gemini-for-Google-Sheet/prompts/#lab-12

Goal

Choose the eight hires that give the highest total Weighted_Score within the client's salary budget, prove the answer, and share only what the client needs.

What You Will Do

The client wants eight hires - two for each of four roles - and one total monthly salary budget of S$41,900. Picking the top two in each role costs S$43,100, which is over budget. You will describe the problem to Gemini's optimisation solver, compare its answer with the no-budget choice, tick the selected candidates and prove the totals with formulas. Then you will export a recommendation to Google Docs, use a Drive file as context, build a minimal Share_View tab and share it safely.

What You Will Build

OrchidBay_Hiring_Decision with a Selected checkbox column, check formulas that prove the constraints, a recommendation in Google Docs and a separate Share_View file shared view-only

Prerequisites

- Lab 11 completed (the rubric Doc is in your Drive), or upload OrchidBay_Screening_Rubric.docx from the Lab 11 folder.
- This lab folder downloaded, including data/OrchidBay_Hiring_Decision.xlsx.

Data note

Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Steps

Step 1. Download OrchidBay_Hiring_Decision.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

PROMPT / CHECK — copy and adapt

```
Required state:
File : OrchidBay_Hiring_Decision (Google Sheets copy)
Tabs : About_This_Data | Finalists (16 rows) | Constraints | Prompt_Log
Budget : Constraints!B8 = S$41,900 per month for 8 hires
```

Step 2. Describe the optimisation problem in plain English: the goal, what can change, and the limits. Gemini's optimisation solver works out the best combination. Note the candidates it picks, their total Weighted_Score and total salary - you prove them in Step 4 and compare them with Test It.

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
I need to choose hires from the Finalists tab. Maximise the total Weighted_Score. Hire exactly
the number of Openings for each Position on the Constraints tab, and keep the total
Expected_Salary of everyone hired at or below the budget in Constraints!B8. Which candidates
should I hire?
```

Step 3. Compare with the obvious choice. Without the budget, the top two per role cost S$43,100 - over the S$41,900 budget - so something has to give. Ask Gemini what it traded and how many points that cost. Note: the optimiser uses Weighted_Score only; remove anyone your Lab 11 review marked Decline before you rely on it.

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
If there were no budget limit, which candidates would you pick, what would they cost, and what
is the trade-off compared with your answer?
```

Step 4. Add the checkbox column with an action, then tick the eight recommended candidates yourself. Paste the checks into Constraints D2:D7 - they must show the budget is met and two hires per role.

PROMPT / CHECK — copy and adapt

```
ACTION:
Add a checkbox column called Selected at the end of the table on the Finalists tab, unchecked
by default.

CHECK:
Total salary =SUMIFS(Finalists!F2:F17,Finalists!M2:M17,TRUE)
Total score =SUMIFS(Finalists!K2:K17,Finalists!M2:M17,TRUE)
Within budget =SUMIFS(Finalists!F2:F17,Finalists!M2:M17,TRUE)<=Constraints!B8
Data Analysts =COUNTIFS(Finalists!C2:C17,"Data Analyst",Finalists!M2:M17,TRUE)
All hires =COUNTIF(Finalists!M2:M17,TRUE)
```

Step 5. Draft the recommendation, check every number against your formulas, then click Export to Docs. Next, use a Drive file as context - expand Sources under the answer to confirm which file Gemini used.

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
Write a hiring recommendation for the client in under 200 words: the eight recommended hires
by role, the total salary against the budget, the one trade-off made because of the budget,
and the risks to check (notice periods and any must-have gaps).

ASK GEMINI:
What are the main points of OrchidBay_Screening_Rubric? Does my recommendation follow its fair
screening rules?
```

Step 6. Share only what the client needs. Ask Gemini which columns are personal data the client doesn't need, then create a minimal view. Do not share this workbook - it still holds names and summaries. Instead right-click the Share_View tab > Copy to > New spreadsheet, open the new file, and share only that file with your trainer as Viewer. Watch for the warning that content may be visible outside your domain.

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
Before I share this spreadsheet with the client, which columns contain personal data they
don't need to see?

ACTION:
Create a new tab called Share_View with only Candidate_ID, Position, Weighted_Score and
Selected for the rows where Selected is checked.
```

Step 7. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

PROMPT / CHECK — copy and adapt

```
LOG:
List the prompts I sent you in this conversation. For each one, give a one-line summary of
what you did, so I can copy it into the Prompt_Log tab.
```

Step 8. Close the loop: give feedback on one helpful and one unhelpful answer with the thumbs up or down. Then, after Prompt_Log is complete, open More options > Clear history to remove the side-panel conversation.

Sample Prompts

Use these prompts in the Gemini side panel in Google Sheets (or adapt them). Name the tab and columns, state the output, and always verify the result. The same prompts are in sample-prompts.txt in this lab folder.

Copy-ready prompts online: https://tertiarycourses.github.io/C817-Gemini-for-Google-Sheet/prompts/#lab-12

#	Purpose	Prompt
1	Optimise	What is the highest total Weighted_Score possible if the budget is reduced by S$2,000?
2	Optimise	Schedule the eight final interviews across Monday and Tuesday so no interviewer has more than three in a row.
3	Docs	Turn the recommendation into a one-page memo with a table of hires and start dates.
4	Gmail	Catch me up on the latest emails about the Q3 hiring drive.
5	Share	Which tabs in this spreadsheet would be risky to share outside Orchid Bay, and why?

Test It

Check	Expected result
Optimum	CAN-26567, CAN-26608, CAN-26621, CAN-26627, CAN-26683, CAN-26692, CAN-26699, CAN-26711.
Totals	Total score 673.5; total salary S$40,700 (budget S$41,900); Within budget = TRUE.
Per role	Two Selected per role; All hires = 8.
No-budget comparison	674.5 for S$43,100; swap CAN-26622 -> CAN-26627.
Share safely	Recommendation exported to Docs; Share_View has 4 columns and 8 candidate rows (plus a header); only the copied Share_View file is shared, as Viewer.

Checkpoint

This is the final lab. Your Drive now holds the full Orchid Bay workflow from raw export to decision.

Troubleshooting

If this happens	Fix
Gemini answers without solving - it just picks the top scores.	Restate the constraints as rules ('exactly 2 per Position', 'total salary at or below S$...'), or ask: 'Use the optimisation solver.' The solver may not be available on every plan - then check the answer with the CHECK formulas.
Gemini's answer breaks the budget.	Tick its picks and look at Within budget - it will show FALSE. Tell Gemini the total and ask again.
The SUMIFS show 0.	The Selected column is not M, or the boxes are not ticked. Check the column letter in the formula.
No optimisation solver on your plan.	Find it by hand: ask Gemini to 'list every pair of candidates for each Position with their combined Weighted_Score and Expected_Salary'. Start from the top pair in each role and swap the pair that loses the fewest points per S$ saved until the total salary fits the budget.
Gemini can't find the rubric file.	Upload OrchidBay_Screening_Rubric.docx from the Lab 11 folder to Drive and save it as a Google Doc.

Challenge

Ask Gemini for the best selection if the client adds a third Data Analyst opening and S$5,000 to the budget. Prove it with the checks.

Reflection

The optimiser maximised a score. What does the score NOT capture that a hiring manager must still judge?